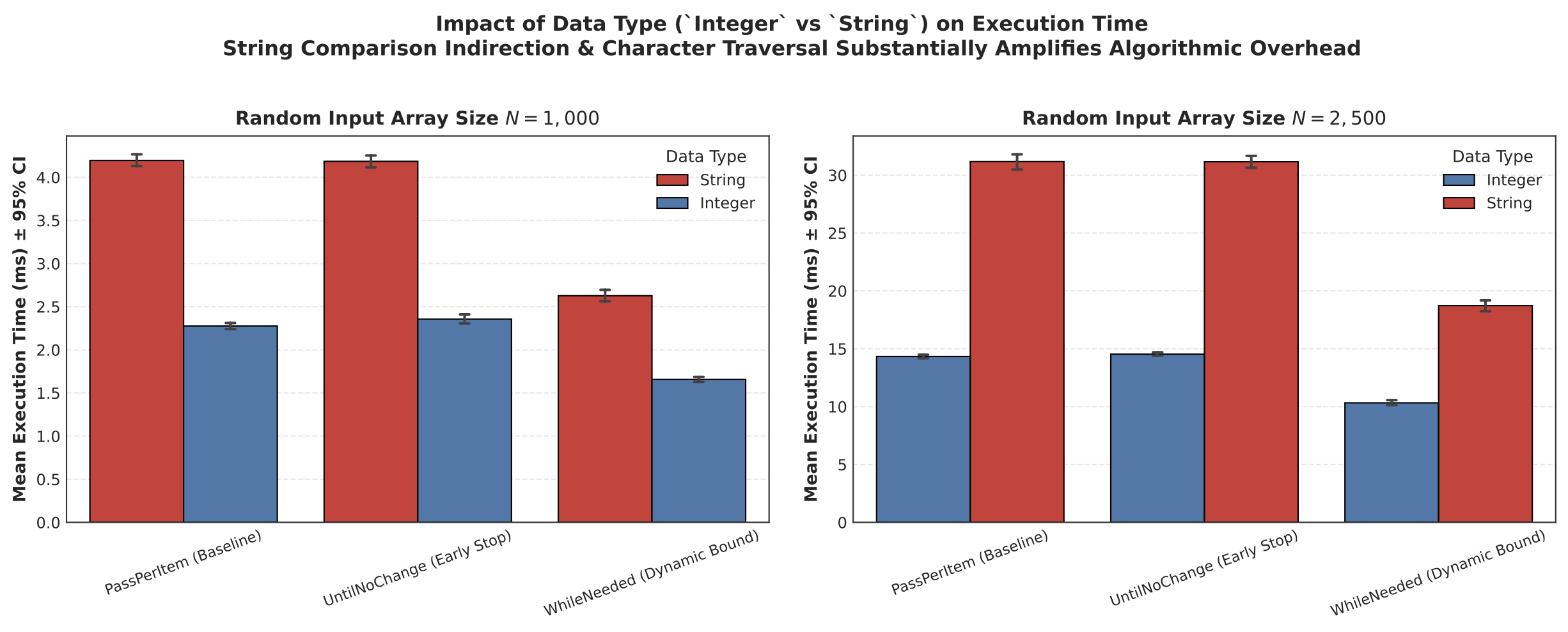
Impact of Data Type (`Integer` vs `String`) on Execution Time
String Comparison Indirection & Character Traversal Substantially Amplifies Algorithmic Overhead
Random Input Array Size $N = 1,000$
Mean Execution Time (ms) ± 95% CI
Data Type
String
Integer
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
PassPerItem (Baseline)
UntilNoChange (Early Stop)
WhileNeeded (Dynamic Bound)
Random Input Array Size $N = 2,500$
Mean Execution Time (ms) ± 95% CI
Data Type
Integer
String
0
5
10
15
20
25
30
PassPerItem (Baseline)
UntilNoChange (Early Stop)
WhileNeeded (Dynamic Bound)